

A Semantic Communication Scheme for Distributed Holographic-Type Communications With Multi-View Images

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Abstract—Holographic-type communication (HTC) is emerging as an important application scenario for the next-generation communication systems. In particular, to provide immersive user experiences, HTC requires to implement 3D visual reconstruction of a physical object by collecting multi-view images from multiple viewpoints. However, it is a dilemma to balance the efficiency and the visual quality of this distributed HTC scenario. To tackle this problem, a distributed multi-view image semantic communication (DMISC) scheme is proposed in this letter. First, a framework consisting of multiple individual semantic encoders and a joint multi-view semantic decoder is proposed to achieve high-quality image recovery by leveraging the correlations among multi-views. Second, a convolutional neural network-based DMISC model is designed to achieve high communication efficiency and visual quality for 3D reconstruction, by combining joint source-channel coding with joint multi-view decoding. Finally, the simulation results are presented to show that our proposed DMISC scheme can achieve better performance than those existing schemes.

Index Terms—Semantic communication, holographic-type communication, multi-view image transmission.

I. INTRODUCTION

WITH the advancement of extended reality applications and artificial intelligence (AI) techniques, holographic-type communication (HTC) is expected to provide users with real-time and immersive experiences in virtual spaces, by transmitting the representations of remote 3D physical scenes [1], [2], [3]. The emerging neural radiance field (NeRF) in [4] techniques can synthesize photo-realistic novel views by reconstructing a radiance field from dense multi-view images, which has significant potential to support the high-quality 3D reconstruction required for HTC systems [5].

In NeRF-based HTC scenarios, dense multi-view images are needed to be transmitted from the collection side to the reconstruction side over complex wireless channels with limited bandwidth. However, traditional communication schemes were designed for accurate bit stream transmission via separate source and channel coding. This leads to low transmission efficiency and poor fault tolerance, making it difficult to

satisfy the low-latency, high-reliability, and high visual quality requirements of HTC scenarios under limited bandwidth [6].

The emerging semantic communication paradigm has the potential to tackle this issue [7], [8]. In [9], the paradigm of semantic communications is first introduced to implement HTC. In [10], a deep learning (DL)-enabled joint source-channel coding (JSCC) scheme has been proposed for more efficient and robust transmission of single-view images. In [11], a DL-based source coding method has been proposed to leverage the correlations of multi-views through a joint context transfer (JCT) module, demonstrating higher recovery quality.

However, it is still challenging to provide sophisticated design of semantic communication transceivers, to achieve high-quality 3D reconstruction with multi-view images, which are transmitted over capacity-limited wireless channels. Motivated by providing some useful insights, a distributed multi-view image semantic communication (DMISC) scheme is proposed in this letter. The contributions are summarized as follows:

- First, a framework comprising multiple individual semantic encoders and a joint multi-view semantic decoder is proposed for multi-view image transmission in distributed HTC scenarios. The key feature of this new framework is that the decoder leverages the correlations among multi-views to achieve high-quality recovery.
- Second, a convolutional neural network (CNN)-based DMISC model is designed to achieve efficient and robust transmission over noisy channels by introducing JSCC, and high visual quality for 3D reconstruction by introducing JCT modules at the decoder.
- Finally, the simulation results are provided to demonstrate that our scheme can achieve better visual quality of multi-view image recovery and 3D reconstruction than the existing schemes under the same communication cost.

II. SYSTEM MODEL

As shown in Fig. 1, we consider a distributed HTC scenario, which consists of N transmitters, denoted by $T_1, \dots, T_N$, and a receiver R . It aims to reconstruct a specific 3D scene $\mathcal{X}$ via the collaborations between the transmitters and the receiver. In particular, the 3D scene reconstruction tasks of the distributed HTC can be accomplished as follows:

- **Local 2D Image Collection:** Each transmitter is equipped with a camera to collect an image from its viewpoint, i.e., an image $\mathbf{x}_n$ is collected by the n -th transmitter T_n .
- **Multi-View Images Transceiving:** After local 2D image collection, all the collected multi-view images

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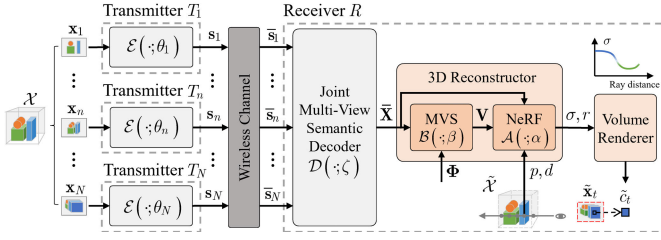


Fig. 1. The framework of our proposed DMISC scheme.

$\mathbf{x}_1, \dots, \mathbf{x}_N$ from the multiple viewpoints are transmitted to the receiver R individually via wireless channels.

- **3D Scene Reconstruction and Rendering for HTC:** Based on the collected multi-view images, a reconstructed and rendered version of the target 3D scene $\mathcal{X}$ is generated by R , to satisfy the HTC service requirements.

In this letter, a semantic communication-based scheme is proposed. Its key idea is to implement JSCC by introducing DL-based encoding and decoding for semantic communications, which has great potential to achieve a sophisticated trade-off between the efficiency and the performance. The details of our proposed scheme are introduced as follows.

A. Semantic Communication-Based Transceivers for Multi-View Images Transmissions

As illustrated in Fig. 1, by employing semantic communication-based transceivers at $T_1, \dots, T_N$ and R , JSCC can be utilized to achieve joint processing gains.

1) *The Semantic Communication Encoder at T_n :* For a specific transmitter T_n , a semantic communication encoder is deployed to implement individual encoding, i.e.,

$$\mathbf{s}_n = \mathcal{E}(\mathbf{x}_n; \theta_n), \quad (1)$$

where $\mathbf{x}_n$ denotes the original collected image, $\mathbf{x}_n \in \mathbb{R}^{L \times 1}$, θ_n denotes the parameters of the semantic encoder $\mathcal{E}(\cdot; \theta_n)$ at T_n , and $\mathbf{s}_n$ is the semantic encoded vector of $\mathbf{x}_n$, $\mathbf{s}_n \in \mathbb{C}^{K \times 1}$. $\mathbf{x}_n$ is a 2D RGB image, L is the number of pixels, i.e., $L = 3 \times H \times W$, where H and W denote the height and the width of $\mathbf{x}_n$, respectively. Moreover, $\mathbf{s}_n$ is a complex-value encoded vector to adapt to the radio frequency signal propagation in wireless circumstances, and K denotes its length. Then, the channel bandwidth ratio (CBR) can be written as K/L , to capture both compression efficiency and bandwidth consumption [10].

2) *The Semantic Communication Decoder at R :* Each encoded message is sent via an additive white Gaussian noise (AWGN) channel, and the observation $\bar{\mathbf{s}}_n$ can be expressed as

$$\bar{\mathbf{s}}_n = \mathbf{s}_n + \mathbf{w}_n, \quad (2)$$

where $\mathbf{w}_n$ denotes the complex AWGN vector, i.e., $\mathbf{w}_n \sim \mathcal{CN}(0, \sigma_i^2 \mathbf{I}_K)$, σ_i^2 is the noise power.

Then, the received encoded messages are jointly recovered by a single semantic decoder employed at R . Since $\bar{\mathbf{s}}_1, \dots, \bar{\mathbf{s}}_N$ are the received multi-view images of a common target 3D scene, there exists semantic redundancy among them, which can provide a great potential of achieving high-quality image recovery with efficient image compression by employing jointly decoding, i.e.,

$$\bar{\mathbf{X}} = \{\bar{\mathbf{x}}_1, \dots, \bar{\mathbf{x}}_N\} = \mathcal{D}(\bar{\mathbf{s}}_1, \dots, \bar{\mathbf{s}}_N; \zeta), \quad (3)$$

where $\bar{\mathbf{X}}$ denotes the joint decoding results of $\mathbf{x}_1, \dots, \mathbf{x}_N$, $\bar{\mathbf{X}} \in \mathbb{R}^{N \times L \times 1}$, $\bar{\mathbf{x}}_n$ denotes the decoded message of $\mathbf{x}_n$, $\bar{\mathbf{x}}_n \in \mathbb{R}^{L \times 1}$, ζ is the model parameters of the semantic decoder $\mathcal{D}(\cdot, \zeta)$.

B. 3D Scene Reconstruction and Rendering for HTC

After decoding the multi-view images, a reconstructed 3D scene $\tilde{\mathcal{X}}$ can be generated by a reconstructor deployed at R . Then, a renderer is used to synthesize a 2D novel view $\tilde{\mathbf{x}}_t$ at a target viewpoint from $\tilde{\mathcal{X}}$ for visualization, and the MVSNeRF scheme in [12] is employed in this letter.

1) *MVSNeRF-Based Implicit Reconstruction:* As shown in Fig. 1, a MVS network $\mathcal{B}(\cdot; \beta)$ is leveraged to reconstruct the neural encoding volume $\mathbf{V}$ from the decoding images $\bar{\mathbf{X}}$, i.e.,

$$\mathbf{V} = \mathcal{B}(\bar{\mathbf{X}}, \Phi; \beta), \quad (4)$$

where β is the model parameters of $\mathcal{B}(\cdot; \beta)$, Φ is the set of the camera parameters of the collected images $\mathbf{x}_1, \dots, \mathbf{x}_N$.

Then, the NeRF network $\mathcal{A}(\cdot; \alpha)$ is used to regress volume rendering properties, including the density σ and the view-dependent radiance r , which can form a radiance field. For a specific 3D coordinate point p and a viewing direction d , the corresponding σ and r can be expressed as

$$(\sigma, r) = \mathcal{A}(p, d, f, \mathbf{c}; \alpha), \quad (5)$$

where f is the neural feature obtained by trilinear interpolation of the neural encoding volume $\mathbf{V}$ at p , i.e., $f = \mathbf{V}(p)$, $\mathbf{c}$ is a $3N$ -channel vector of the pixel colors of all of the decoding images, i.e., $\mathbf{c} = [\bar{\mathbf{x}}_n(u_n, v_n)]$, and the pixel location (u_n, v_n) is calculated by projecting p onto the view $\bar{\mathbf{x}}_n$.

2) *Volume Rendering:* The design of the volume renderer is based on the volume rendering equation with radiance field in [4] and [12], the RGB color of a target pixel $\tilde{\mathbf{c}}_t$ of $\tilde{\mathbf{x}}_t$ is calculated by marching a ray through the pixel and accumulating (σ, r) at the sampled shading points on it, i.e.,

$$\tilde{\mathbf{c}}_t = \sum_m \exp\left(-\sum_{j=1}^{m-1} \sigma_j\right) (1 - \exp(-\sigma_m)) r_m. \quad (6)$$

III. THE PROPOSED CNN-BASED DMISC SCHEME

In this section, the model structure of the proposed CNN-based DMISC model is designed to implement semantic encoding and decoding. Then, the loss function and training strategy are introduced.

A. The Architecture of Semantic Encoder $\mathcal{E}(\cdot; \theta_n)$ in (1) at T_n

Similar to DeepJSCC in [10], a CNN-based semantic encoder is employed to perform joint source and channel encoding, which can map image pixels into transmission symbols directly. As shown in Fig. 2, the equipped encoders are identical at all the transmitters, which consists of two convolution blocks CB_1 and CB_2 to extract the encoded tensor-version semantic features $\mathbf{y}_n$ of original image $\mathbf{x}_n$, and a symbolization block SB to map $\mathbf{y}_n$ into transmit symbols.

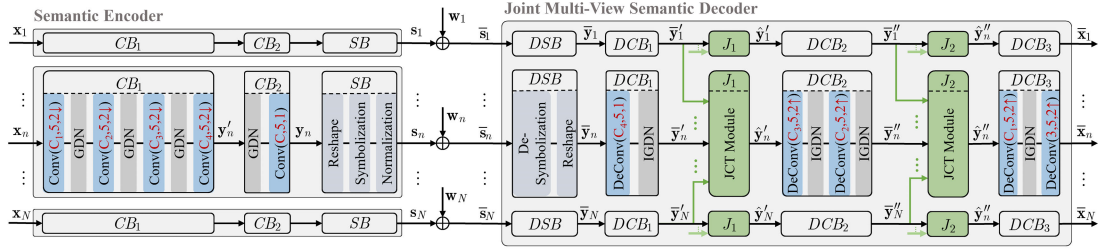


Fig. 2. The architecture of the proposed CNN-based DMISC model. (The convolution layers are parameterized by (c, l, s) , where c is the number of channel outputs, s is the stride, l is the length of the square filter, and $\uparrow$ and $\downarrow$ indicate up- and down-sampling, respectively.)

1) *The Architecture of CB_1 and CB_2* : The intermediate feature y'_n of x_n is first encoded by CB_1 , $y'_n \in \mathbb{R}^{C_4 \times \frac{H}{16} \times \frac{W}{16}}$. CB_1 consists of four down-sampling convolution layers, where the stride is set as 2, and the last channel output size is denoted as C_4 . Then, the semantic coded tensor y_n is generated via compressing y'_n by CB_2 , $y_n \in \mathbb{R}^{C \times \frac{H}{16} \times \frac{W}{16}}$, and C is the number of channel outputs of CB_2 . Please note that the generalized divisive normalization (GDN) layers are used to normalize the latent features, which can guarantee the stability and the convergence of model training.

2) *The Architecture of SB* : y_n is first reshaped into a $\frac{CHW}{256} \times 1$ vector. Then, it is symbolized into a one-dimensional complex symbol vector s'_n , $s'_n \in \mathbb{C}^{\frac{CHW}{512} \times 1}$, where each complex element is formed by pairing two real numbers. Finally, s'_n is normalized to follow the transmit power constraint, i.e., $s_n = s'_n / \mathbb{E}(\|s'_n\|_2)$. The CBR of $\mathcal{E}(\cdot; \theta_n)$ is managed by adjusting C of CB_2 to balance the efficiency and the visual quality of semantic encoding on x_n .

B. The Architecture of Semantic Decoder $\mathcal{D}(\cdot; \zeta)$ in (3) at R

In this part, a semantic decoder $\mathcal{D}(\cdot; \zeta)$ is designed to jointly decode all the received images. As shown in Fig. 2, a de-symbolization block DSB and three de-convolution blocks (DCBs), i.e., DCB_1 , DCB_2 , and DCB_3 , are employed to provide basic semantic decoding capability for each image, which is similar to the conventional semantic decoder in [10] by reversing the operations of semantic encoder $\mathcal{E}(\cdot; \theta_n)$ in Section III-A. To leverage the semantic correlation of multi-view images to achieve high-quality decoding, two JCT modules J_1 and J_2 are deployed between the DCB modules across the output features of different images. The detailed design is introduced as follows.

1) *The Architectures of Basic Semantic Decoding Modules DSB , DCB_1 , DCB_2 , and DCB_3* : The main tasks of DSB , DCB_1 , DCB_2 , and DCB_3 are to decode the received images individually in parallel, which provides the basic semantic decoding functions. In particular, each received message $\bar{s}_n$ is first mapped into a real-value semantic coded tensor $\bar{y}_n$ via DSB , $\bar{y}_n \in \mathbb{R}^{C \times \frac{H}{16} \times \frac{W}{16}}$. Then, the DCB modules DCB_1 , DCB_2 , and DCB_3 are employed to extract the convoluted features, which execute the reversed operations of CB modules at the semantic encoder.

2) *The Architecture of JCT Modules J_1 and J_2* : To enhance the decoding performance, the JCT modules are used to fully explore the semantic correlated features of multi-view images [11]. For a specific received image $\bar{s}_n$ the JCT module J_1 is deployed between DCB_1 and DCB_2 to fuse the

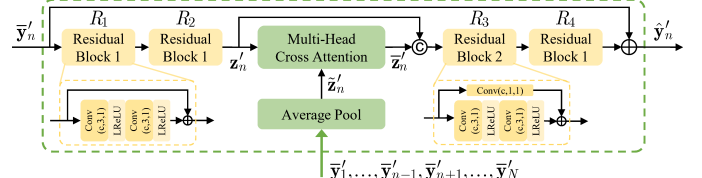


Fig. 3. The illustration of the first JCT module J_1 .

intermediate features $\bar{y}'_1, \dots, \bar{y}'_{n-1}, \bar{y}'_{n+1}, \dots, \bar{y}'_N$ from other related views, to enrich its own feature $\bar{y}'_n$. As shown in Fig. 3, the semantic feature of $\bar{y}'_n$ is extracted by two residual blocks R_1 and R_2 , which is denoted as z'_n . Meanwhile, the intermediate features of other multi-view images are fused by using an average pool, and the average result is $\bar{z}'_n = \frac{1}{N-1} \sum_{i=1, i \neq n}^N \bar{y}'_i$. Then, a multi-head cross-attention block is applied to explore the correlations between z'_n and $\bar{z}'_n$, i.e.,

$$\bar{z}'_n = \text{Conv}(\mathbf{A}_{n,1} \oplus \dots \oplus \mathbf{A}_{n,j} \oplus \dots \oplus \mathbf{A}_{n,h}),$$

$$\text{and } \mathbf{A}_{n,j} = \sigma_{\text{row}}(\mathbf{Q}_{n,j}) \left(\sigma_{\text{col}}(\mathbf{K}_{n,j})^T \mathbf{V}_{n,j} \right), \quad (7)$$

where $\sigma_{\text{row}}(\mathbf{B})/\sigma_{\text{col}}(\mathbf{B})$ denotes the softmax function along each row/column of a matrix $\mathbf{B}$, $\oplus$ is the channel-wise concatenation, h is the number of heads, $\mathbf{Q}_{n,j} = \text{Conv}_j(z'_n)$ denotes the query of a convolution layer, and $\mathbf{K}_{n,j} = \text{Conv}_j(\bar{z}'_n)$ and $\mathbf{V}_{n,j} = \text{Conv}_j(\bar{z}'_n)$ denote its key and value, respectively. Finally, another set of residual blocks R_3 and R_4 are used to refine the semantic features after the multi-view fusion to obtain $\hat{y}'_n$.

Similar to J_1 , J_2 is used to fuse the intermediate features provided by DCB_2 , and its architecture is identical to J_1 .

C. Loss Function and Training Strategy

To minimize the difference between the recovered and the original multi-view images with a fixed CBR, the mean squared error (MSE) is used as the loss function, i.e.,

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^N \text{MSE}(\mathbf{x}, \bar{\mathbf{x}}) = \frac{1}{N} \sum_{n=1}^N \left(\frac{1}{L} \|\mathbf{x} - \bar{\mathbf{x}}\|^2 \right). \quad (8)$$

In Algorithm 1, a two-stage training strategy is designed. During Stage 1, to avoid the risks of error propagation across multi-view images, the JCT modules of $\mathcal{D}(\cdot; \zeta)$ are blocked, which enables that the training of semantic encoder and decoder for the transmissions of each image can be performed individually. It can ensure that our proposed scheme can initially converge to achieve acceptable performance. During Stage 2, to fully explore the semantic correlations to improve

Algorithm 1 The Two-Stage Training Strategy of DMISC

- 1: **Initialization:** The model parameters $\theta^{(0)}$ and $\zeta^{(0)}$, the numbers of iterations E_1 and E_2 for Stage 1 and 2, respectively.
- 2: **Stage 1:** J_1 and J_2 in $\mathcal{D}(\cdot; \zeta)$ are blocked, then the semantic decoder can be constructed as $\hat{\mathcal{D}}(\cdot; \zeta) = \{DSB \rightarrow DCB_1 \rightarrow DCB_2 \rightarrow DCB_3\}$, for the e_1 -th iteration, $1 \leq e_1 \leq E_1$
 - The semantic encoder and decoder of each single-view image can be trained individually, i.e., $\mathcal{E}(\cdot; \theta^{(e_1-1)})$ and $\hat{\mathcal{D}}(\cdot; \zeta^{(e_1-1)})$ are trained in tandem by calculating $\mathcal{L}$ in (8).
- 3: **Stage 2:** J_1 and J_2 are added back to participate the training of semantic decoder $\mathcal{D}(\cdot; \zeta)$. For the e_2 -th iteration, $1 < e_2 \leq E_2$
 - Update $\mathbf{s}_1, \dots, \mathbf{s}_N$ individually by $\mathcal{E}(\cdot; \theta^{(e_2-1)})$.
 - Update $\bar{\mathbf{x}}_1, \dots, \bar{\mathbf{x}}_N$ jointly by $\mathcal{D}(\cdot; \zeta^{(e_2-1)})$.
 - Calculate $\mathcal{L}$ to update $\theta^{(e_2)}$ and $\zeta^{(e_2)}$.
- 4: **Output:** The final training results of $\mathcal{E}(\cdot; \theta)$, $\mathcal{D}(\cdot; \zeta)$.

the recovery performance of semantic decoding, the JCT modules are added back and participate in the joint training.

IV. SIMULATION RESULTS

In this section, the performance of our proposed DMISC scheme is evaluated. The number of transmitters is set as $N = 3$. The DTU dataset in [13] is adopted, which consists of a training set with 109 scenes and a testing set with 8 scenes. The original images are with 512×640 pixels, which are randomly flipped and cropped into 256×256 patches for data augmentation during training, to enhance generalization ability of the model to new data. The MVSNeRF-based 3D reconstruction and rendering models are pre-trained by following [12]. During the training procedure, the Adam optimizer is employed, the initial learning rate is set as 10^{-4} , the batch size is set as 8, the numbers of epochs of both training stages are set as $E_1 = E_2 = 200$, and the average SNR is 4 dB.

The following performance metrics are evaluated:

- Peak signal-to-noise ratio (PSNR): It is a pixel-wise metric, which measures the ratio between the maximum possible value of the image pixels and the MSE.
- Multi-scale structural similarity (MS-SSIM) in [14]: It is a structural perceptual metric, which measures the similarity of two images at multiple scales by calculating luminance, contrast, and structure.
- Learned perceptual image patch similarity (LPIPS) in [15]: It is a DL-enabled perceptual metric, which can imitate the human perceptual assessment process to give the LPIPS loss score. LPIPS is more aligned with human visual perception than PSNR and MS-SSIM [15].

Moreover, the following typical transmission schemes are selected as baselines:

- Traditional scheme¹: The better portable graphics (BPG) codec in [16] and the low-density parity-check code

¹This scheme is labeled as “BPG+LDPC+ M -QAM” in the provided figures.

TABLE I
THE COMPUTATIONAL COSTS OF DIFFERENT SCHEMES

Scheme		Computational Latency (ms)		CBR
		Encoder	Decoder	
Traditional scheme	BPG	558.4	165.4	0.0645
	LDPC+QAM	11.3	1584.5	
SISC		5.5	4.3	0.0625
DMISC		5.7	15.6	0.0625

(LDPC) are employed for source and channel coding, respectively, followed by M -QAM modulation.

- Single-view image semantic communication (SISC) scheme: Similar to [10], each single-view image is decoded individually without the JCT modules in Section III-B.
- Upper bound: The original images are perfectly sent to the receiver without any loss caused by communications.

Fig. 4(a), Fig. 4(b), and Fig. 4(c) depict the recovery performance of different schemes using PSNR, MS-SSIM, and LPIPS, respectively. As shown in these figures, the transmission robustness of our proposed DMISC scheme is significantly improved compared to the traditional schemes, which can show the benefits of employing JSCC in semantic communications. In particular, a single model trained at an SNR of 4 dB can effectively cover a range of SNR from 0 to 10 dB, and the recovery performance can be improved as the SNR increases. In contrast, the traditional separated source and channel coding methods suffer from the “cliff effect”, i.e., the recovery performance decreases sharply in the lower SNR region due to decoding failure, and fails to improve in the high SNR region. Meanwhile, our proposed DMISC scheme can achieve better recovery performance, especially for the perceptual metrics MS-SSIM and LPIPS. It demonstrates that the CNN-based DMISC transceivers can capture subtle image details, i.e., textures and edges, due to its strong capacity for nonlinear feature learning. Furthermore, our proposed DMISC scheme with joint multi-view decoding can achieve better recovery performance than the SISC scheme with individual decoding.

In Fig. 4(d), the LPIPS performance of rendering is provided. It shows that our DMISC scheme significantly outperforms the traditional communication schemes. Moreover, our proposed DMISC scheme achieves better performance than the SISC scheme in the high SNR region, and approaches the upper bound. As shown in Fig. 4(c) and Fig. 4(d), our proposed DMISC scheme shows superior performance in terms of noise robustness and image recovery compared to the baselines. This ultimately leads to improved 3D reconstruction and rendering performance. The simulation results verify the benefits of integrating the JSCC and JCT modules.

Fig. 5 depicts the rate-distortion results, where the values of SNR are set as 4 dB and 10 dB, respectively. The results show that our proposed DMISC scheme can always achieve the best LPIPS performance. It verifies that our proposed DMISC scheme can significantly improve the transmission efficiency and the visual quality in distributed HTC scenarios. Fig. 6 shows that our proposed training strategy in Algorithm 1 can achieve better convergence and performance.

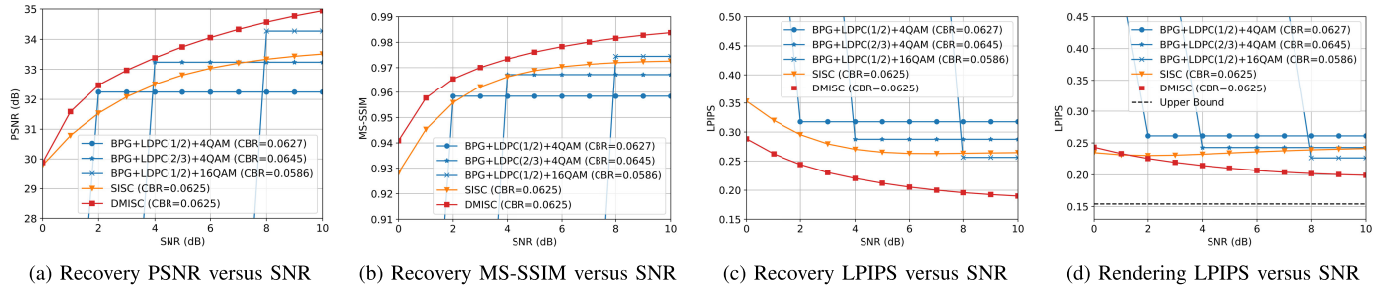


Fig. 4. The performance comparisons of multi-view image recovery and novel view rendering over AWGN channels.

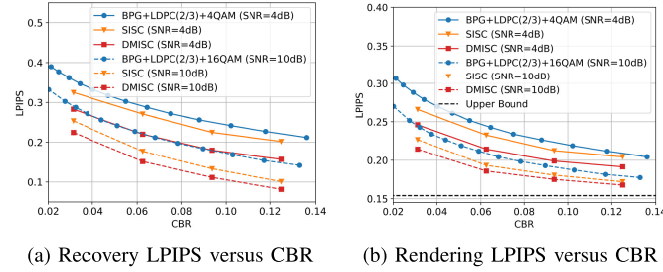


Fig. 5. The LPIPS performance.

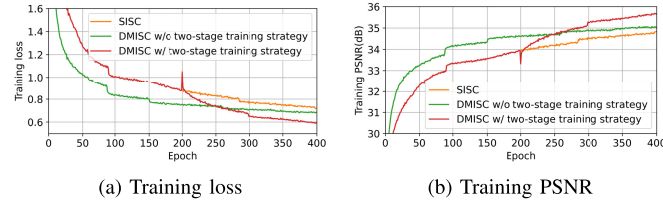


Fig. 6. The training performance results with different training strategies.

In Table I, the computational costs are provided. Both of the SISC and our proposed DMISC schemes are tested by using a single RTX 3090 GPU, as well as the processing of LDPC and QAM for the traditional scheme. Moreover, an Intel Xeon Gold 6242R CPU is employed to support BPG codec of the traditional scheme, and its computational latency is much larger than the semantic communication-based schemes. The parameter sizes of encoder and decoder are both 4.38 M for SISC scheme. The parameter size of semantic decoder of our proposed DMISC scheme is increased to 16.05 M, since two JCT modules are added to enable joint decoding by fusing the intermediate features of multi-view images.² The computational latency of semantic decoding for our proposed DMISC scheme is increased from 4.3 ms to 15.6 ms in equal proportion. Considering the significant performance gains achieved by our proposed DMISC scheme, this increment of computational latency is acceptable, since it is still with the same scale as the SISC scheme, and can be fast executed by using a single mainstream GPU.

V. CONCLUSION

In this letter, a DMISC scheme has been proposed for distributed HTC scenarios. First, a semantic communication framework was proposed to support efficient multi-view image

transmissions for HTC. Second, the CNN-based semantic transceivers are designed to implement joint semantic decoding of multi-view images. Finally, the simulation results verify the effectiveness of our proposed scheme.

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²The parameter sizes of J_1 and J_2 are 8.58 M and 3.09 M, respectively.